LAB 00: Building Your Home Lab & Professional Portfolio Infrastructure

Lab Overview

This lab builds everything that supports Labs 01–14. By the end, you will have a fully functional virtual environment, a help desk ticketing system, a GitHub portfolio, and a Hashnode blog ready to publish. No fictional company — this is **your** infrastructure.

Estimated Total Time: 5–7 hours (split across 2 sessions recommended)

Difficulty: ⭐☆☆☆☆ (long, not hard)

What You'll Build

When Lab 00 is complete, you will have:

```
PC (Host)
│
├── VirtualBox 7.x (Hypervisor)
│   ├── NAT Network: "LabNAT" (10.0.2.0/24 — internet access)
│   ├── Host-Only Network: "LabLAN" (192.168.56.0/24 — internal)
│   │
│   ├── [VM] osTicket-Server (Ubuntu Server 22.04)
│   │   └── Apache + MySQL + PHP + osTicket
│   │   └── IP: 192.168.56.10
│   │   └── Status: RUNNING
│   │
│   ├── [VM] Win10-PC01 (Shell — empty, configured, not installed)
│   │   └── IP: 192.168.56.100 (future)
│   │
│   ├── [VM] Win11-PC01 (Shell — empty, configured, not installed)
│   │   └── IP: 192.168.56.101 (future)
│   │
│   ├── [VM] Ubuntu-PC01 (Shell — empty, configured, not installed)
│   │   └── IP: 192.168.56.102 (future)
│   │
│   └── [VM] Mint-PC01 (Shell — empty, configured, not installed)
│       └── IP: 192.168.56.103 (future)
│
├── Documentation Method (your choice of platform)
```

```
│
│
└── ISOs Folder
    ├── Windows 10 Enterprise Evaluation
    ├── Windows 11 Enterprise Evaluation
    ├── Ubuntu Desktop 22.04.x LTS
    ├── Linux Mint 21.x
    └── Ubuntu Server 22.04.x LTS
```

PART A: VirtualBox Installation & Configuration

Estimated Time: 30 minutes

A1: Download VirtualBox

Open your browser and navigate to:

<https://www.virtualbox.org/wiki/Downloads>

1.
2. Under **VirtualBox platform packages**, click **Windows hosts**

3. The file `VirtualBox-7.x.x-xxxxx-Win.exe` will download (~100MB)

While that downloads, scroll down on the same page to **VirtualBox Extension Pack** — click **All supported platforms** to download:

Oracle_VirtualBox_Extension_Pack-7.x.x.vbox-extpack

4. ***Why the Extension Pack? It adds USB 2.0/3.0 support, disk encryption, PXE boot, and other features we'll need for peripheral and USB labs.***

5. Save both files to your `Downloads` folder

A2: Install VirtualBox

1. Right-click `VirtualBox-7.x.x-xxxxx-Win.exe` → **Run as administrator**
2. Click **Next** on the welcome screen
3. On the **Custom Setup** screen:
    - Leave all features selected (default)

Change the installation path to:

C:\\Program Files\\Oracle\\VirtualBox\\

    -
    - Click **Next**

4. On the **Warning: Network Interfaces** screen:
    - This warns that installation will temporarily reset your network connection
    - Click **Yes** to proceed
5. Click **Install**
6. If prompted by Windows Security, click **Install** to trust Oracle's drivers
7. Check **Start Oracle VirtualBox after installation**
8. Click **Finish**

A3: Install Extension Pack

1. In VirtualBox Manager, go to **File** → **Tools** → **Extension Pack Manager**

   (In VirtualBox 7.0+: you may also go to **File** → **Preferences** → **Extensions**)

2. Click the **Install** button (green "+" icon)

Navigate to your Downloads folder and select:

Oracle_VirtualBox_Extension_Pack-7.x.x.vbox-extpack

3.
4. Read the license agreement, scroll to the bottom, click **I Agree**

5. Wait for installation to complete

6. Verify the Extension Pack shows as **Active** in the list

A4: Configure Global Preferences

1. Go to **File** → **Preferences** (or press `Ctrl+G`)

2. **General** tab:

Default Machine Folder: Change this to a dedicated location:

C:\\VirtualBox_VMs

-
- Create this folder if it doesn't exist

3. ***Why? Keeping VMs off the system drive (if you have multiple drives) or in a predictable location makes backup and management easier. On your 1TB drive, space isn't an issue — organization is.***

4. Click **OK**

Create the ISO storage folder:

C:\\VirtualBox_ISOs

5.

Create a lab documentation folder:

C:\\HomeLab

6.

[📸 SCREENSHOT: Preferences showing your Default Machine Folder]

Verification: Open File Explorer and confirm these three folders exist:

- `C:\\VirtualBox_VMs`
- `C:\\VirtualBox_ISOs`
- `C:\\HomeLab`

PART B: Virtual Network Architecture

Estimated Time: 20 minutes

B1: Understanding VirtualBox Network Modes

Before configuring, here's what each mode does and when we use it:

Mode	Internet Access	VM-to-VM	Host-to-VM	Our Use Case
NAT	✅	❌	❌	Quick internet for single VM
NAT Network	✅	✅	❌	VMs sharing internet access
Host-Only	❌	✅	✅	Simulated office LAN — our primary lab network
Bridged	✅	✅	✅	Not used (exposes VMs to real network)
Internal	❌	✅	❌	Isolated testing (future labs)

Our architecture uses two networks per VM:

- **Adapter 1 (NAT):** Gives each VM internet access for updates, downloads
- **Adapter 2 (Host-Only):** Simulates the office LAN where all machines see each other and the host can access them

---

B2: Create/Configure Host-Only Network

1. In VirtualBox Manager, go to **File** → **Tools** → **Network Manager**

(Or **File** → **Host Network Manager** in some versions)

2. Click the **Create** button to create a new Host-Only Network
    - It will be named `VirtualBox Host-Only Ethernet Adapter` (or `vboxnet0` on some versions)
3. Select the new network and click **Properties**

Configure the **Adapter** tab:

text
IPv4 Address: 192.168.56.1
IPv4 Network Mask: 255.255.255.0

4.
5. Configure the **DHCP Server** tab:
    - **Enable Server:** ✅ Checked

text
Server Address: 192.168.56.100
Server Mask: 255.255.255.0
Lower Bound: 192.168.56.100
Upper Bound: 192.168.56.200

6. ***Why enable DHCP? Client VMs will get addresses automatically during initial setup. We'll assign static IPs later in specific labs. The osTicket server will get a static IP manually.***

7. Click **Apply** then **Close**

Host-Only Network configuration showing IP settings

DHCP Server settings

B3: Verify Host-Only Network on Your Host

1. Open **Command Prompt** (press `Win+R`, type `cmd`, press Enter)

Run:

cmd
ipconfig /all

2.

Look for an adapter named **VirtualBox Host-Only Network** or similar. Confirm:

text
IPv4 Address: 192.168.56.1
Subnet Mask: 255.255.255.0

3.

B4: Network Architecture Documentation

Create a file `C:\\HomeLab\\network-documentation.md` and paste the following:

Markdown
Home Lab Network Documentation

Network Architecture

Host Machine
- Hostname: [YOUR-COMPUTER-NAME]
- Host OS: Windows [10/11]
- Host-Only IP: 192.168.56.1

Networks
Network Name	Type	Subnet	Purpose
Default NAT	NAT	10.0.2.0/24	Internet access for VMs
LabLAN	Host-Only	192.168.56.0/24	Internal lab network

IP Address Assignments
Hostname	Role	IP Address	OS	Status
Host	Physical Machine	192.168.56.1	Windows	Active
osTicket-Server	Help Desk Server	192.168.56.10	Ubuntu Server 22.04	Pending
Win10-PC01	Client Workstation	192.168.56.100	Windows 10	Pending
Win11-PC01	Client Workstation	192.168.56.101	Windows 11	Pending
Ubuntu-PC01	Client Workstation	192.168.56.102	Ubuntu 22.04	Pending

| Mint-PC01 | Client Workstation | 192.168.56.103 | Linux Mint 21 | Pending |

Network Diagram

text

```
                ┌──────────────────┐
                │   INTERNET   │
                └──────────────────┘
                        │
            ┌──────────────────────────┐
            │  NAT (10.0.2.0/24)   │  ← Each VM gets its own NAT
            │  Gateway: 10.0.2.2   │    for internet access
            └──────────────────────────┘
                        │
      ┌─────┬─────┬─────┬
```

STEP 2: If on Windows, it pushes the Media Creation Tool Download and run it

STEP 3: Select "Create installation media for another PC"

STEP 4: Choose: Language: English (United States) Edition: Windows 10 Architecture: 64-bit

STEP 5: Select "ISO file" (not USB)

STEP 6: Save to C:\VirtualBox_ISOs\

NOTE: This gives you Windows 10 Home/Pro, not Enterprise. For a home lab this works fine — the core skills are identical. Enterprise features (BitLocker, Group Policy) are available in Pro.

C2: Windows 11 Enterprise Evaluation

Navigate to:

<https://www.microsoft.com/en-us/evalcenter/evaluate-windows-11-enterprise>

1.
2. Same process as above — download the **64-bit ISO**
3. Save to `C:\\VirtualBox_ISOs\\`

Rename to:

Windows11_Enterprise_Eval.iso

4.

C3: Ubuntu Desktop 22.04 LTS

Navigate to:

<https://releases.ubuntu.com/22.04/>

1.
2. Download the **Desktop image** (ubuntu-22.04.x-desktop-amd64.iso)

3. Save to `C:\\VirtualBox_ISOs\\`

> ***Why 22.04 and not 24.04? Version 22.04 LTS has mature driver support in VirtualBox and guaranteed compatibility with all tools we'll use. You can use 24.04 if you prefer.***

C4: Linux Mint 21

Navigate to:

<https://www.linuxmint.com/download.php>

1.
2. Select **Cinnamon Edition** (most user-friendly desktop)
3. Choose any mirror and download
4. Save to `C:\\VirtualBox_ISOs\\`

C5: Ubuntu Server 22.04 LTS

Navigate to:

<https://releases.ubuntu.com/22.04/>

1.
2. Download the **Server install image** (ubuntu-22.04.x-live-server-amd64.iso)
3. Save to `C:\\VirtualBox_ISOs\\`

C6: Verify Your ISO Collection

Open `C:\\VirtualBox_ISOs\\` and confirm you have 5 files:

```
C:\\VirtualBox_ISOs\\
├── Windows10_Enterprise_Eval.iso       (~5 GB)
├── Windows11_Enterprise_Eval.iso       (~5 GB)
├── ubuntu-22.04.x-desktop-amd64.iso    (~3.6 GB)
├── linuxmint-21.x-cinnamon-64bit.iso   (~2.7 GB)
└── ubuntu-22.04.x-live-server-amd64.iso (~1.8 GB)
```

PART D: osTicket Server Build

Estimated Time: 90 minutes

This is the most involved section of Lab 00. Take your time — this server supports every future lab.

D1: Create the Ubuntu Server VM

1. In VirtualBox Manager, click **New** (or `Ctrl+N`)

Name and Operating System:

```
Name:      osTicket-Server
Folder:    C:\\VirtualBox_VMs (should be default)
ISO Image: C:\\VirtualBox_ISOs\\ubuntu-22.04.x-live-server-amd64.iso
Type:      Linux
Version:   Ubuntu (64-bit)
```

2.
   - **Uncheck** "Skip Unattended Installation" if it appears checked — actually we WANT to **skip** the unattended install, so make sure **Unattended Installation is disabled/skipped**
3. ***In VirtualBox 7.0, if you attach an ISO and it detects a supported OS, it may try to do an unattended install. We want manual control. Check the box that says "Skip Unattended Installation."***

4. Click **Next**

Hardware:

Base Memory: 2048 MB (2 GB)
Processors: 1 CPU

5.
6. Click **Next**

Virtual Hard Disk:

Create a Virtual Hard Disk Now: Selected
Disk Size: 20 GB

7.
    - Leave "Pre-allocate Full Size" **unchecked** (dynamic allocation saves space)
8. Click **Next**, review the summary, click **Finish**

D2: Configure VM Network Adapters

Before booting, configure the network:

1. Select **osTicket-Server** in the VM list → click **Settings**

Go to **Network**:

Adapter 1:

Enable Network Adapter: ✅
Attached to: NAT

2. **Adapter 2:**
    - Click the **Adapter 2** tab

Enable Network Adapter: ✅
Attached to: Host-only Adapter
Name: VirtualBox Host-Only Ethernet Adapter

3.
4. Click **OK**

Adapter 1 (NAT) settings

Adapter 2 (Host-Only) settings

D3:

Install Ubuntu Server 22.04

1. Select **osTicket-Server** and click **Start**

2. The VM boots from the ISO. You'll see the Ubuntu Server installer.

3. **Language:** Select **English** → press Enter

4. **Installer Update:** If prompted to update the installer, select **Continue without updating** (faster, avoids issues)

Keyboard Configuration:

Layout: English (US)
Variant: English (US)

5. Select **Done**

6. **Choose type of install:** Select **Ubuntu Server** (not minimized)

7. **Network Connections:**

    - You should see two interfaces:
        - `enp0s3` — connected via DHCPv4 (this is your NAT adapter)
        - `enp0s8` — may show "not configured" (this is your Host-Only adapter)
    - **Leave defaults for now** — we'll configure the static IP after installation
    - Select **Done**
8. **Configure Proxy:** Leave blank → **Done**

9. **Ubuntu Archive Mirror:** Leave default → **Done**

10. **Guided Storage Configuration:**

    - Select **Use an Entire Disk**
    - **Leave LVM unchecked** (simpler for our lab)
    - Select **Done**
11. **Storage Configuration Summary:** Review → Select **Done**

    - Confirm destructive action → **Continue**

Profile Setup:

Your name: Lab Admin
Your server's name: osticket-server
Pick a username: labadmin
Choose a password: **********
Confirm password: **********

12. ***Important: Write these credentials down. You'll use them constantly.***

13. **Upgrade to Ubuntu Pro:** Select **Skip for now** → **Continue**

14. **SSH Setup:**

    - **Install OpenSSH Server:** ✅ Check this box
    - This allows you to SSH from your host, which is more comfortable than the VM console
    - Select **Done**
15. **Featured Server Snaps:** Don't select anything → **Done**

16. Wait for installation to complete (5-10 minutes)

17. When you see **"Installation complete!"**, select **Reboot Now**

18. If prompted "Please remove installation medium," just press **Enter** (VirtualBox handles this automatically)

Wait for the server to boot. Log in with:

Username: labadmin
Password: *******

19.

[📸 SCREENSHOT: Successful login to Ubuntu Server terminal]

D4: Post-Install System Configuration

Update the system:

sudo apt update &&sudo apt upgrade -y

1. Enter your password when prompted. This may take several minutes.

Configure the static IP on the Host-Only adapter:

First, identify your network interfaces:

ip addr show

2. You should see:
    - `lo` — loopback
    - `enp0s3` — NAT adapter (has a 10.0.2.x address)
    - `enp0s8` — Host-Only adapter (may have a 192.168.56.x DHCP address or no address)

Edit the Netplan configuration:

sudo nano /etc/netplan/00-installer-config.yaml

3.

Replace the contents with (be careful with YAML spacing — use exactly 2 spaces for each indent level, **no tabs**):

YAML
```
network:
  ethernets:
    enp0s3:
      dhcp4:true
    enp0s8:
```

addresses:
-192.168.56.10/24
version:2

4. ***What this does:*** ← ***FIXED*** ◦ ***`enp0s3`*** ***(NAT): Gets IP automatically for internet access*** ◦ ***`enp0s8`*** ***(Host-Only): Gets static IP 192.168.56.10 — this is how your host machine will reach osTicket***

5. **Save and exit:** Press **`Ctrl+O`**, then **`Enter`**, then **`Ctrl+X`**

Apply the network configuration:

```
sudo netplan apply
```

6.

Verify the static IP:

```
ip addr show enp0s8
```

You should see:

```
inet 192.168.56.10/24
```

7.

Verify internet connectivity:

```
ping -c 4 google.com
```

8. You should see 4 successful replies.

Test connectivity from your HOST machine. Open Command Prompt on your Alienware:

```
ping 192.168.56.10
```

9. You should see successful replies. If not, verify the Host-Only network is configured correctly in Part B.

D5: Install Apache Web Server

Back in the Ubuntu Server terminal:

Install Apache:

```
sudo apt install apache2 -y
```

1.

Enable and start Apache:

```
sudo systemctl enable apache2
sudo systemctl start apache2
```

2.

Verify Apache is running:

```
sudo systemctl status apache2
```

3. Look for `Active: active (running)` in green.

Test from your HOST browser. Open a browser on your Alienware and navigate to:

<http://192.168.56.10>

4. You should see the **Apache2 Ubuntu Default Page** — a page that says "It works!"

D6: Install MySQL Database Server

Install MySQL:

```
sudo apt install mysql-server -y
```

1.

Enable and start MySQL:

```
sudo systemctl enable mysql
sudo systemctl start mysql
```

2.

Secure the MySQL installation:

```
sudo mysql_secure_installation
```

Answer the prompts:

```
VALIDATE PASSWORD COMPONENT:          n
New password:                         LabMySQL2024!
Re-enter new password:                ************
Remove anonymous users:               y
Disallow root login remotely:         y
Remove test database:                 y
Reload privilege tables:              y
```

3.

Create the osTicket database and user:

```
sudo mysql
```

At the `mysql>` prompt, enter each line followed by Enter:

```
CREATE DATABASE osticket;
CREATE USER 'osticket_user'@'localhost' IDENTIFIEDBY 'osTicketDB2024!';
GRANT ALL PRIVILEGESON osticket.* TO 'osticket_user'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

4.

D7: Install PHP and Required Extensions

Install PHP and all extensions required by osTicket:

```
sudo apt install php php-mysql php-cgi php-fpm php-cli php-curl php-gd php-imap php-mbstring
php-intl php-apcu php-xml php-zip -y
```

1.

Verify PHP installation:

```
php -v
```

2. You should see PHP 8.1.x.

Enable Apache mod_rewrite (required for osTicket):

```
sudo a2enmod rewrite
sudo systemctl restart apache2
```

3.

D8: Download and Install osTicket

Install unzip utility:

```
sudo apt install unzip -y
```

1.

Download osTicket (check [**https://osticket.com/download/**](https://osticket.com/download/) for the latest version — using v1.18.1 as reference):

```
cd /tmp
wget <https://github.com/osTicket/osTicket/releases/download/v1.18.1/osTicket-v1.18.1.zip>
```

2. ***If this URL doesn't work: Go to `https://github.com/osTicket/osTicket/releases` in your host browser, find the latest release, right-click the .zip download link, copy the URL, and paste it into the wget command.***

Extract osTicket:

```
unzip osTicket-v1.18.1.zip -d /tmp/osticket
```

3.

Move the upload directory to Apache's web root:

```
sudo mv /tmp/osticket/upload /var/www/html/osticket
```

4.

Create the configuration file from the sample:

```
sudo cp /var/www/html/osticket/include/ost-sampleconfig.php
/var/www/html/osticket/include/ost-config.php
```

5.

Set proper permissions:

```
sudo chown -R www-data:www-data /var/www/html/osticket
sudo chmod 0666 /var/www/html/osticket/include/ost-config.php
```

6.

D9: Configure osTicket via Web Installer

On your HOST machine, open a browser and navigate to:

<http://192.168.56.10/osticket/setup/>

1.
2. You should see the **osTicket Installer** page showing prerequisites. Verify that most items show green checkmarks. If `php-imap` shows a warning, it's non-critical for our lab.

3. Click **Continue**

System Settings:

Helpdesk Name: HomeLab IT Support
Default Email: helpdesk@homelab.local

4.

Admin User:

First Name: Lab
Last Name: Admin
Email Address: admin@homelab.local
Username: your choice
Password: *********
Confirm Password: *********

5.

Database Settings:

MySQL Table Prefix: ost_
MySQL Hostname: localhost
MySQL Database: osticket
MySQL Username: osticket_user
MySQL Password: **********

6.
7. Click **Install Now**

8. You should see **"Congratulations!"** — osTicket has been installed.

Important post-installation steps. Back in the Ubuntu Server terminal:

Lock down the config file:

sudo chmod 0644 /var/www/html/osticket/include/ost-config.php
Remove the setup directory:

sudo rm -rf /var/www/html/osticket/setup

1.

Bookmark these URLs on your HOST browser:

Staff Panel: <http://192.168.56.10/osticket/scp/>
Client Portal: <http://192.168.56.10/osticket/>

2.

Log in to the Staff Panel:

text
URL: <http://192.168.56.10/osticket/scp/>
Username: labadmin
Password: *********

3.

D10: Configure osTicket for the Lab Series

Now let's set up the ticketing structure that mirrors a real IT department.

SLA Plans

1. In the Staff Panel, go to **Admin Panel → Manage → SLA**

2. Click **Add New SLA Plan** and create four plans:

Name	Grace Period	Schedule
SEV-A (Critical)	1 hour	24/7
SEV-B (High)	4 hours	24/5

SEV-C (Normal)	8 hours	Monday–Friday 8am–5pm
SEV-D (Low)	24 hours	Monday–Friday 8am–5pm

Departments

1. Go to **Admin Panel** → **Agents** → **Departments**

2. A default "Support" department exists. Add:

Department	Type
IT Support	Public
System Administration	Public
Networking	Public

Help Topics

1. Go to **Admin Panel** → **Manage** → **Help Topics**

2. Create the following:

Help Topic	Department	SLA
Password Reset	IT Support	SEV-D
Hardware Issue	IT Support	SEV-C
Software Issue	IT Support	SEV-C
Network Connectivity	Networking	SEV-B
Access / Permissions	IT Support	SEV-C
Workstation Not Booting	IT Support	SEV-B
Malware / Security Incident	System Administration	SEV-A
New Employee Setup	IT Support	SEV-C
Printer Issue	IT Support	SEV-D
General Inquiry	IT Support	SEV-D

Teams

1. Go to **Admin Panel** → **Agents** → **Teams**
2. Create:
    - **Tier I Support**
    - **Tier II Support**

Agent (Your Account)

1. Go to **Admin Panel** → **Agents** → **Add New Agent**

Create your technician persona:

```
 text
First Name:    [YOUR FIRST NAME]
Last Name:     [YOUR LAST NAME]
Email:         tech@homelab.local
Username:      tech
```

2.
    - Set a password
    - Assign to **IT Support** department
    - Add to **Tier I Support** team
    - Set **Access** to All departments with ability to create and edit tickets

---

D11: Take a VM Snapshot

This is **critical**. Before doing anything else, snapshot your working osTicket server.

1. In VirtualBox Manager, select **osTicket-Server**

2. Click the **Snapshots** tab (hamburger menu → Snapshots, or the camera icon)

3. Click **Take** (camera icon)

Name it:

```
Name:        Clean-osTicket-Install
Description:  Fresh Ubuntu Server 22.04 + LAMP + osTicket configured.
              All SLAs, departments, help topics, and teams created.
```

Static IP: 192.168.56.10

4.
5. Click **OK**

> ***Why snapshot? If anything breaks in future labs, you can restore to this clean state in seconds instead of rebuilding from scratch. This mirrors real-world practices: snapshot before changes, rollback if needed.***

D12: Credential Reference Sheet

Create a file `C:\\HomeLab\\credentials.md`:

```
Markdown
# Home Lab Credentials
⚠️ LAB USE ONLY — Never use these passwords in production

## Ubuntu Server (osTicket-Server)
- IP: 192.168.56.10
- SSH User: labadmin
- SSH Pass: ********

## MySQL
- Root: (socket auth, use sudo mysql)
- DB Name: osticket
- DB User: osticket_user
- DB Pass: **********

## osTicket Staff Panel
- URL: <http://192.168.56.10/osticket/scp/>
- Admin User: labadmin
- Admin Pass: **********
- Tech User: tech
- Tech Pass: [whatever you set]

## osTicket Client Portal
- URL: <http://192.168.56.10/osticket/>
```

> ***Security Note: In a real environment, you'd use a password manager. For a local-only lab, this file is acceptable. Do NOT push this file to your public GitHub repo — we'll add it to `.gitignore`.***

PART E: Client VM Shell Creation

Estimated Time: 30 minutes

These VMs are **configured but not installed**. The OS installations happen in Labs 01 and 02. Right now we're just creating the empty VMs with the right specs.

E1: Windows 10 VM Shell

1. Click **New** in VirtualBox Manager

Configure:

Name: Win10-PC01
Folder: C:\\VirtualBox_VMs
ISO Image: (leave empty — we'll attach it in Lab 02)
Type: Microsoft Windows
Version: Windows 10 (64-bit)
Skip Unattended: ✅

2.

Hardware:

Base Memory: 4096 MB (4 GB)
Processors: 2

3.

Virtual Hard Disk:

Create a Virtual Hard Disk Now
Disk Size: 50 GB
Pre-allocate: unchecked

4.

5. Click **Finish**

Select **Win10-PC01** → **Settings** → **Network:**

Adapter 1: NAT (enabled)
Adapter 2: Host-only Adapter (enabled)

6.
7. Click **OK**

E2: Windows 11 VM Shell

1. Click **New**

Configure:

Name: Win11-PC01
Folder: C:\\VirtualBox_VMs
ISO Image: (leave empty)
Type: Microsoft Windows
Version: Windows 11 (64-bit)
Skip Unattended: ✅

2. ***VirtualBox 7.0+ automatically enables EFI and TPM 2.0 when you select Windows 11 (64-bit). If your version doesn't, we'll set it manually in step 6.***

Hardware:

Base Memory: 4096 MB (4 GB)
Processors: 2

3.

Virtual Hard Disk:

Disk Size: 64 GB
Pre-allocate: unchecked

4.
5. Click **Finish**

6. **Verify Windows 11 requirements.** Select **Win11-PC01** → **Settings:**

    - **System** → **Motherboard:** Confirm **Enable EFI** is ✅ checked
    - **System** → **Motherboard:** Confirm **TPM** dropdown shows **v2.0** (VirtualBox 7.0+)
7. If you don't see a TPM option:

    - Go to **Settings** → **Security** (in some VB versions) or **Settings** → **System** → **Encryption**
    - Enable **TPM 2.0**
8. ***If your VirtualBox version doesn't support TPM: You can still install Windows 11 using registry workarounds during installation. We'll cover this in Lab 02 if needed.***

Network:

Adapter 1: NAT (enabled)
Adapter 2: Host-only Adapter (enabled)

9.

E3: Ubuntu Desktop VM Shell

1. Click **New**

Configure:

Name: Ubuntu-PC01
Folder: C:\\VirtualBox_VMs
ISO Image: (leave empty)
Type: Linux
Version: Ubuntu (64-bit)
Skip Unattended: ✅

2.

Hardware:

Base Memory: 4096 MB (4 GB)
Processors: 2

3.

Virtual Hard Disk:

Disk Size: 25 GB
Pre-allocate: unchecked

4.

Network:

Adapter 1: NAT (enabled)
Adapter 2: Host-only Adapter (enabled)

5.

E4: Linux Mint VM Shell

1. Click **New**

Configure:

Name: Mint-PC01
Folder: C:\\VirtualBox_VMs
ISO Image: (leave empty)
Type: Linux
Version: Ubuntu (64-bit)
Skip Unattended: ✅

2. ***Why "Ubuntu" for Linux Mint? Mint is Ubuntu-based. VirtualBox doesn't have a Mint-specific option. Selecting Ubuntu (64-bit) provides the correct hardware defaults.***

Hardware:

Base Memory: 2048 MB (2 GB)
Processors: 2

3.

Virtual Hard Disk:

Disk Size: 25 GB
Pre-allocate: unchecked

4.

Network:

Adapter 1: NAT (enabled)
Adapter 2: Host-only Adapter (enabled)

5.

E5: Verify All VMs

Your VirtualBox Manager should now show 5 VMs:

text
```
VirtualBox Manager
├── osTicket-Server     [Running or Saved] 🟢
├── Win10-PC01          [Powered Off] ⚫
├── Win11-PC01          [Powered Off] ⚫
├── Ubuntu-PC01         [Powered Off] ⚫
└── Mint-PC01           [Powered Off] ⚫
```

E6: Storage Audit

Let's verify disk usage. Open File Explorer, navigate to `C:\\VirtualBox_VMs\\` and check the total size:

```text
osTicket-Server/   ~4-5 GB (OS installed)
Win10-PC01/        ~few MB (empty shell)
Win11-PC01/        ~few MB (empty shell)
Ubuntu-PC01/       ~few MB (empty shell)
Mint-PC01/         ~few MB (empty shell)
```

Total current usage should be under 10 GB. As OSes are installed, this will grow.

PART F: Documenting Your Work

Estimated Time: Varies by platform

F1: Why Documentation Matters

Building this lab is an accomplishment. But if you can't show what you built and explain what you learned, it's invisible to employers, peers, and your future self.

The technical skills from this lab — virtualization, networking, Linux server administration, web application deployment — are valuable. But the ability to **document what you did, why you did it, and what the outcome was** is what separates a hobbyist from a professional.

Every IT department runs on documentation. Every ticket requires resolution notes. Every project needs a summary. If you build the habit now, it becomes second nature in the workplace.

F2: Choose Your Platform

How you document is less important than **that** you document. Choose whatever platform fits your workflow:

Platform	Best For	Cost
GitHub / GitLab	Code-focused portfolios, version control, technical audiences	Free

Hashnode / Dev.to	Blog-style write-ups, community engagement, SEO visibility	Free
Medium	Long-form writing, broader non-technical audience	Free (limited)
WordPress	Full website control, custom branding	Free (with hosting costs for custom domain)
Notion	Personal knowledge base, internal documentation	Free
LinkedIn Articles	Direct employer visibility, professional networking	Free
Physical Notebook	Personal reference, interview prep	Cost of notebook

> ***My recommendation: Use at least two — one for detailed technical documentation (like GitHub or a blog) and one for professional visibility (like LinkedIn). The detailed version proves depth. The professional version gets you noticed.***

Whatever you choose, be consistent. A portfolio with 15 documented labs on any platform is more impressive than a perfectly designed website with 2 posts.

F3: Use the STAR Method

Every lab you complete should be documented using the **STAR method** — the same framework used in behavioral job interviews. This serves two purposes: it structures your portfolio AND it prepares you to talk about your work in interviews.

STAR stands for:

S — SITUATION

What was the context? Who needed help? What was the business impact? What made it urgent?

Example: "NovaBridge Technologies, a managed service provider, signed three new clients. Field technicians needed standardized bootable USB toolkits before onboarding site visits the following week."

T — TASK

What were YOU specifically responsible for? What were the constraints — deadline, tools available, scope?

Example: "As the IT Support Specialist, I needed to create three types of bootable media and write documentation so any technician could recreate the toolkit independently."

A — ACTION
What did you actually DO? Be specific — name the tools, the commands, the steps. Include what DIDN'T work and how you pivoted. This is where your technical depth shows.

Example: "I used Rufus to create a Windows 11 GPT bootable USB, Ventoy for a multi-boot Linux diagnostic drive, and the built-in Windows Recovery Drive tool. During testing, the Windows USB failed to boot on a UEFI machine. I diagnosed the root cause using diskpart, identified an MBR/GPT mismatch, and recreated the USB with the correct partition scheme."

R — RESULT
What was the measurable outcome? How many users were impacted? How much time was saved? What did you prevent from happening again?

Example: "All three toolkits were delivered 2 days before deadline. The Ventoy USB consolidated 4 tools into one drive, reducing equipment per technician by 75%. The documentation I created was adopted as the standard procedure, preventing future boot failures."

STAR Template

Use this template for every lab you complete:

STAR Writeup — Lab [XX]: [TITLE]

Situation

[WHO] at [COMPANY] experienced [PROBLEM] which caused [BUSINESS IMPACT]. [URGENCY/CONTEXT details].

Task

As [YOUR ROLE], I needed to [OBJECTIVE] while ensuring [CONSTRAINT]. The deadline was [TIMEFRAME] and [ADDITIONAL REQUIREMENTS].

Action

1. **Initial Assessment:** [What I checked first and what I observed]
2. **Diagnosis:** [What tools/commands I used and what I discovered]
3. **Resolution:** [Specific technical steps I took to fix the issue]
4. **Verification:** [How I confirmed the fix worked]
5. **Prevention:** [What I documented or recommended to prevent recurrence]

What Didn't Work (and Why)

[Describe the first approach that failed and what it taught you about the root cause — this shows troubleshooting depth]

Result

- ✅ [Primary outcome with specific metric]
- ✅ [Secondary outcome — users impacted, time saved]
- ✅ [Preventive measure implemented]
- 📝 Documented in ticketing system with full root cause analysis

> ***Why include "What Didn't Work"? Because troubleshootingis never a straight line. Showing your first attempt faileddemonstrates that you can adapt, learn, and persist.Interviewers specifically look for this. Anyone can describea success — professionals can describe the failures that ledto the success.***

F4: What to Document for Each Lab

At minimum, every lab should produce:

1. WHAT YOU BUILT / FIXED
    - Screenshots or screen recordings of key moments
    - Before and after states
    - The tools and commands you used

2. WHAT YOU LEARNED
    - Key concepts (in your own words, not textbook definitions)
    - What surprised you
    - What you'd do differently next time
3. HOW IT CONNECTS TO THE JOB
    - Which IT roles use this skill daily
    - How you'd explain this to a non-technical manager
    - What interview question this lab prepares you for
4. THE TICKET
    - If you built the osTicket system: log every lab as a real ticket with resolution notes
    - If you didn't: write resolution notes anyway in whatever format you choose
    - The habit of documenting resolutions is more valuable than the platform you use

> ***The goal is not perfection. The goal is consistency. A roughwrite-up published every week beats a polished masterpiecepublished never. Start messy. Get better as you go. The firstpost is always the hardest — every one after that gets easier.***

F5: Show Your Work

However you choose to document, make it **visible**. A portfolio

that only exists on your local machine helps nobody — including you.

Some ways to show your work:

- **Post weekly on LinkedIn** — even a 5-sentence summary of whatyou learned this week builds your professional presence
- **Share in IT communities** — Simply Cyber, r/ITCareerQuestions,r/CompTIA, Discord servers, local meetups
- **Reference it in job applications** — link to your portfolio inyour resume, cover letter, and LinkedIn profile
- **Talk about it in interviews** — "Let me show you the lab wherel built this" is more powerful than "I studied this for the exam"

You built something real. Make sure people can see it.

For [lab-00-condensed.md]

Replace the deleted Parts F and G with:

Part F: Document Your Work

Choose a documentation platform (GitHub, blog, Notion, LinkedIn, etc.)

Document each lab using the STAR method:

```text
S — Situation:  What was the context and business impact?
T — Task:       What were you responsible for?
A — Action:     What did you do? (tools, commands, steps)
R — Result:     What was the measurable outcome?
```

Minimum output per lab:

1. Screenshots of key moments (before/after)
2. Tools and commands used
3. What you learned (your words, not textbook)
4. Ticket resolution notes (osTicket or equivalent)
5. STAR writeup for interview prep

Make it visible:

Post weekly. Share in communities.

Link in applications. Talk about it in interviews.

Updated Verification Checklist

Replace the deleted GitHub and Hashnode checklist sections with:

For lab-00-full.md

```markdown
### **Documentation**

```text
[ ] Chosen a documentation platform (blog, GitHub, Notion, etc.)
[ ] Understand the STAR method and can apply it to each lab
[ ] Have a plan for making your work visible (LinkedIn, community, etc.)
[ ] Credentials stored securely (not publicly accessible)
[ ] Network documentation created with IP table and diagram

---
```
```

Updated Verification Checklist

Replace the deleted GitHub and Hashnode checklist sections with:

For lab-00-full.md

```markdown
### **Documentation**

```text
[ ] Chosen a documentation platform (blog, GitHub, Notion, etc.)
[ ] Understand the STAR method and can apply it to each lab
[ ] Have a plan for making your work visible (LinkedIn, community, etc.)
[ ] Credentials stored securely (not publicly accessible)
[ ] Network documentation created with IP table and diagram

## Final Checklist

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Go through this checklist and confirm every item. Check each box as you verify.

### **VirtualBox Environment**


text [ ] VirtualBox installed and opens without errors [ ] Extension Pack installed and shows "Active" [ ] Default Machine Folder set to C:\VirtualBox_VMs [ ] Host-Only Network configured (192.168.56.1/24) [ ] Host-Only DHCP server enabled (range .100-.200)


### **Virtual Machines**


text [ ] osTicket-Server — Running, accessible via SSH and browser [ ] Win10-PC01 — Created, powered off, dual NICs configured [ ] Win11-PC01 — Created, powered off, EFI + TPM enabled, dual NICs [ ] Ubuntu-PC01 — Created, powered off, dual NICs configured [ ] Mint-PC01 — Created, powered off, dual NICs configured


### **osTicket**

text [ ] Apache serves pages at http://192.168.56.10 [ ] osTicket Staff Panel accessible at http://192.168.56.10/osticket/scp/ [ ] Can log in as labadmin [ ] Can log in as tech [ ] 4 SLA plans created (SEV-A through SEV-D) [ ] 3 departments created (IT Support, System Admin, Networking) [ ] 10 help topics created [ ] 2 teams created (Tier I, Tier II) [ ] Snapshot "Clean-osTicket-Install" saved

### **Network**

text [ ] Host can ping 192.168.56.10 (osTicket server) [ ] osTicket server can ping google.com (internet access) [ ] Network documentation file created with IP table and diagram [ ] Credentials file created (NOT in GitHub repo)

### **ISOs**

text [ ] Windows 10 Enterprise Evaluation ISO downloaded [ ] Windows 11 Enterprise Evaluation ISO downloaded [ ] Ubuntu Desktop 22.04 LTS ISO downloaded [ ] Linux Mint 21 ISO downloaded [ ] Ubuntu Server 22.04 LTS ISO downloaded [ ] All ISOs stored in C:\VirtualBox_ISOs\

---

## **🎯 Skills From Lab 00**

Even though this is a setup lab, you practiced real skills:

| **Skill** | **What You Did** |
| --- | --- |
| **System Configuration** | Configured VirtualBox preferences, VM hardware specs, EFI/TPM settings |
| **Local Area Networks** | Designed and implemented a virtual network with NAT and Host-Only adapters, assigned static IPs |
| **Data Storage Technologies** | Created dynamically allocated virtual hard disks, planned storage allocation |
| **Command-Line Interface** | Configured Ubuntu Server via CLI, installed LAMP stack |
| **Information Systems Security** | Secured MySQL, set file permissions, created credential documentation, configured .gitignore for sensitive files |
| **System Support** | Built and configured a Linux server from scratch, installed and configured a web application |

---

## ** 💡 What You'd Tell an Employer**

> ***"Before starting any labs, I designed and built the entire infrastructure from scratch — a virtualized network with five machines, a help desk ticketing system running on a LAMP stack, and a documentation process for tracking every lab with measurable outcomes. Every lab I completed was logged as a real ticket with SLA tracking."***
>

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## **⏭️ Bridge to Lab 01**

Your environment is built. Your portfolio infrastructure is live. Your ticketing system is ready to receive its first ticket.

In **Lab 01**, you step into the role of a technician at **NovaBridge Technologies**, a managed service provider. Your manager needs a field toolkit: bootable Windows and Linux USBs for client site visits. You'll create bootable media using Rufus and Ventoy — and when one of them won't boot, you'll troubleshoot why.

---

## **Quick Reference Card**

Save this somewhere accessible — you'll reference it constantly:

text

```
╔═══════════════════════════════════════════════════════════════════════╗
║ HOME LAB QUICK REFERENCE ╒
╟───────────────────────────────────────────────────────────────────────╢
╒ ╒ ╒ osTicket Staff Panel: ╒ ╒ http://192.168.56.10/osticket/scp/ ╒ ╒ User: labadmin /
osTicket2024! ╒ ╒ ╒ ╒ osTicket Server SSH: ╒ ╒ ssh labadmin@192.168.56.10 ╒ ╒ Pass:
LabAdmin2024! ╒ ╒ ╒ ╒ IP Addresses: ╒ ╒ Host: 192.168.56.1 ╒ ╒ osTicket: 192.168.56.10 ╒
╒ Win10: 192.168.56.100 ╒ ╒ Win11: 192.168.56.101 ╒ ╒ Ubuntu: 192.168.56.102 ╒ ╒ Mint:
192.168.56.103 ╒ ╒ ╒ ╒ ╒
╚═══════════════════════════════════════════════════════════════════════╝
```

##